

[A Specific, Researchable Collaborative Proposal Title]

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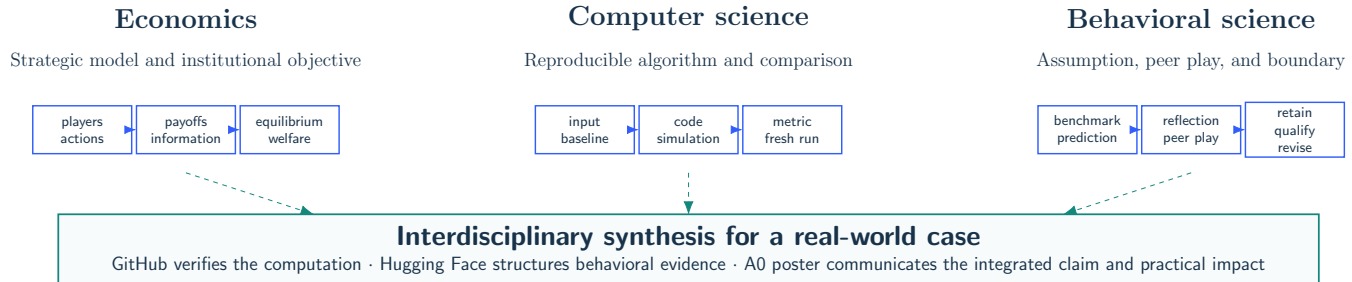


Figure 1: One integrated PS2 evidence chain. Economic reasoning defines the strategic question and institutional objective; reproducible computation tests the model; behavioral evidence probes assumptions; the A0 poster communicates the combined claim and its practical implications for a concrete real-world case. Replace every placeholder and distinguish completed evidence from planned work.

Submission to: COMSCI/ECON 206 — Computational Microeconomics.

Course session: Autumn 2026 Session 1. **Team:** [FPXX].

Symposium: [A/B/C/D].

Course instructor: Professor Luyao Zhang. **Assignment:** PS2 collaborative research proposal.

PS1-continuous template. This file intentionally retains the PS1 ACM layout, five-section main paper, Author Notes, references, and Appendices A–E. Appendix F contains only the additional PS2 team, auction, linked-artifact, poster, and symposium records. Keep Sections 1–5 within two main pages, including metadata, Figure 1, its caption, and artifact links.

1 Three connected research questions and verified gap

State the background, motivation, application, affected stakeholders, and strategic tension. Compare the closest verified work and identify the gap that remains.

Q1—Economics: [Which incentive, welfare, allocation, or institutional question needs an answer?]

Q2—Computation: [Which model, algorithm, simulation, or evaluation makes the answer checkable?]

Q3—Behavior: [Which belief, preference, framing effect, reasoning limit, or learning process could change the prediction?]

End with one shared question that requires all three answers. Explain how Figure 1 represents the connection. Team members may contribute differently, but every author must be able to explain the integrated argument.

2 Answer to Q1: incentives and strategic model

State a provisional answer or falsifiable hypothesis. Define players, feasible actions and complete strategies, objectives or payoffs, timing, information, and beliefs where needed. Select and cite the appropriate solution concept—for example Nash equilibrium [2], subgame-perfect equilibrium, or Bayesian equilibrium [1]. Translate the same problem through game theory, social choice, and mechanism design. Explain what each lens changes rather than merely naming it.

3 Answer to Q2: computation, auction comparison, and artifacts

Describe inputs, language-independent pseudocode, baseline, one meaningful modification, metrics, and one fresh-run result. For the auction or allocation application, identify the scarce resource, participants, values or signals, bidding or reporting rule, allocation and payment rules, stopping rule, and desired outcome. Compare two mechanisms or two information/behavioral assumptions; state where the auction analogy fails.

Computational scientist—GitHub: [Your team GitHub repository and exact commit/release]

Runnable notebook: [Your runnable notebook URL]

Behavioral scientist—Hugging Face: [Your behavioral-science Hugging Face repository or Space]

Interdisciplinary scholar—A0 poster: [Your editable A0 poster filename or stable URL]

Report the exact commit/release, notebook cell or command, parameters, output, and interpretation. Label every untested extension as planned.

4 Answer to Q3: behavior and interdisciplinary integration

State the rational benchmark, a behavioral mechanism, and a competing explanation. Use self-reflection or structured peer play as exploratory evidence, not population-level validation. Explain how the Hugging Face artifact structures the decision setting and evidence. Specify what observation would distinguish the explanations and how it would make the team retain, qualify, or revise an economic assumption, algorithm, or interface.

5 Advanced development, real-world impact, and roadmap

Build on named Nobel- and Turing-recognized foundations where relevant. Identify what is distinctive about the present environment, what current research leaves unresolved, and what becomes knowable or designable only through the integration of economics, computation, behavioral evidence, and public communication.

For one concrete real-world case, identify affected stakeholders, feasible benefits, risks, whose knowledge is represented, what evidence would establish value, and how the system can be questioned or corrected. Compare the evidence standards of academia, industry, and public institutions. State the next discriminating test.

Table 1: From foundations to a collaborative interdisciplinary contribution.

Horizon	Milestone and evidence
Foundation	[Defining work, enduring question, and retained insight.]
PS2	[Strategic model, linked artifacts, auction comparison, and verified result.]
Symposium	[A0-poster claim, real-world case, peer challenge, and response.]
Next study	[Discriminating test, broader validation, and safeguard.]
2056	[Integrated contribution and evidence supporting the aspiration.]

Expected 2056 Nobel Prize and Turing Award contribution statement. By 2056, we aspire to be recognized for [theory, method, or design] addressing [enduring human question]. Building on [verified foundations], we integrate [economic insight], [computational contribution], and [behavioral evidence] to advance [specific boundary]. Progress will be demonstrated by [milestones and evidence], with practical value for [people and institutions] and accountable correction through [mechanism].

Open Science Statement

[Identify the GitHub and Hugging Face repositories, code/data or synthetic inputs, dependencies, licenses, exact commit/release, shortest reproducible run, actual outputs, and limitations. The editable A0 poster and exported PDF must point to the same evidence.]

Statement of Contribution to the UN’s SDGs

[Name the relevant SDG target only where justified. State the intended mechanism, audience, access needs, observable indicator, possible unintended effects, and evaluation plan. Do not present intended benefit as demonstrated impact.]

Submission milestones

Review-ready version: Sunday, September 27, 2026, 11:00 P.M. Beijing time. Symposium presentation, peer evaluation, and practical-impact discussion: Monday, September 28, 2026, 2:45–5:15 P.M., IB 2050. Final revised submission: Wednesday, September 30, 2026, 11:00 P.M. Beijing time, before the October 1–2 national holiday.

Author Notes

Acknowledgements

Course instructor: Professor Luyao Zhang. We thank [full names of peers, discussants, community or industry partners] for [specific contribution actually used]. We acknowledge [data, software, adapted materials, and permitted AI assistance] accurately. Feedback or a program role does not automatically imply coauthorship. The authors remain responsible for the proposal and remaining errors.

Team and individual contributions

[State the team's joint intellectual contribution in 100–150 words. Then repeat for each author: full name and author order; personal intellectual decision; inspectable output and location; verification and result; connection to another part of the joint work; next responsibility after symposium feedback. Avoid labels such as "coding" or "writing" without an intellectual decision and evidence.]

References

- [1] John C. Harsanyi. 1968. Games with Incomplete Information Played by Bayesian Players, II: Bayesian Equilibrium Points. *Management Science* 14, 5 (1968), 320–334. doi:10.1287/mnsc.14.5.320
- [2] John F. Nash. 1950. Equilibrium Points in n-Person Games. *Proceedings of the National Academy of Sciences* 36, 1 (1950), 48–49. doi:10.1073/pnas.36.1.48

A Technical details and reproducibility

[Supply derivations, pseudocode, inputs, package versions, seed where relevant, baseline, comparison, metrics, actual outputs, known failures, and limitations. Identify the GitHub commit, notebook cells, Hugging Face version, A0 poster version, and fresh-run record.]

A.1 AI-use disclosure

AI-Assisted drafting and debugging are permitted only within the assignment designation after initial independent reasoning. Record tool/model, date, purpose, significant prompts, accepted or rejected suggestions, human changes, and verification. If none, state "No generative AI was used." The human authors are responsible for every claim, citation, and computation.

B Cumulative Intellectual Development

[Retain the PS1 pattern: earlier claim → dated evidence or feedback → revision → intellectual consequence. Explain how PS1 or Week 4 work developed into the PS2 shared question without forcing topic continuity. Name the affected section, figure, repository location, or abstract row.]

C Open-source and three-lens inspiration

[Preserve the PS1 source record where relevant. For each source or tool, state the official URL, license, version/date, capability, limitation, and design consequence. Record how game theory, social choice, and mechanism design alter the

shared model. Distinguish actual observation from interpretation and planned evidence.]

D Review response and revision record

[Retain the PS1 response pattern and add PS2 team review. Preserve review-ready v1, two peer reviews per student, discussant/instructor feedback where received, author decisions, changed locations, rerun results, and final v2. Use Keep, Revise, or Investigate and justify the decision.]

E Structured abstract and cumulative proposal map

Update the same eight-row cumulative map used in PS1. Replace every prompt with a brief, concrete entry and preserve the prior version.

Table 2: Appendix E structured abstract template; retain the familiar PS1 structure.

Proposal element	Current statement / change
Background	State the strategic setting, established knowledge, unresolved issue, and key foundation.
Motivation and application	Explain why the issue matters now, the concrete case, affected people, and need.
Three connected questions	State economics, computation, behavior, and the integrated question that requires all three.
Method and model assumptions	Specify actors, incentives, timing, information, strategic model, baseline, computation, and evidence.
Results or expected results	State the actual result and evidence type; separately state the planned test and falsifier.
Intellectual merits	Explain what is learned beyond the cited foundation and why integration changes the answer.
Practical impacts	Identify private/public value, access, risks, evidence needed, correction route, and justified SDG mechanism.
Feedback received and revision made	Name the dated source, earlier claim, decision, actual revision, verification, and consequence.

Keep detailed technical records, literature notes, open-source inspirations, acknowledgements, and AI-use disclosure in the supporting sections. The main paper, linked artifacts, poster, and structured abstract must use the same question, assumptions, and evidence status.

F PS2 extension record: auction, linked artifacts, poster, and symposium

F.1 Auction laboratory record

[For each treatment, record the game URL or identifier, settings, value or signal, planned bid, equilibrium prediction, actual bid/outcome, payment, utility, revenue, allocation efficiency, behavioral observation, and resulting revision. State the allocation, payment, and stopping rules explicitly.]

F.2 Linked-artifact parity

Artifact	Required parity check
GitHub	[Same model, inputs, commit, runnable path, output, and limits.]
Hugging Face	[Same decision setting, behavioral assumption, evidence status, and privacy boundary.]
A0 poster	[Same research question, result, real-world case, limitations, instructor, acknowledgements, and links.]

F.3 Symposium and practical-impact record

[Retain the September 28 presentation version, peer-evaluation forms, questions, and the team’s Keep/Revise/Investigate decisions. For the real-world case, record the affected stakeholders, proposed benefit, risk, missing evidence, and correction route. Identify every change made for the September 30 final revision and rerun the relevant check.]